- We can find w_0, w_1 that give min $J(w_0, w_1)$ by taking partial derivative of $J(w)$ with respect to w_0, w_1 seperately.

Approch1 Ordinary linear regression (OLS)

A closed form solution in terms of linear regression, refer to an analytical method to find the exact coefficient of polynomial by minimizing the error without iterative optimization.

$$\frac{\partial}{\partial w_0} J(w) = 0 \qquad \frac{\partial}{\partial w_1} J(w) = 0$$

$$\begin{cases} w_0 = \bar{y} - \hat{w}_1 \bar{x} = \dfrac{\sum_{i=1}^{m} y_i - \sum_{i=1}^{m} \hat{w}_1 x_i}{m} \\ \hat{w}_1 = \dfrac{cov(x,y)}{s_x^2} = r_{xy} \dfrac{s_y}{s_x} \end{cases}$$

- There is a relation between slope and correlation coefficient
- There is a relation between slope and s_x and s_y.

→ Self-study for derivation

Multiple linear regression

One dependent variable (Y)

Multiple independent variable $(x_1, x_2, \cdots x_n)$

n = Dimension / feature

m = number of training samples.

$$y_i = w_0 + w_1 x_{i1} + w_2 x_{i2} + \cdots + w_n x_{in} + \varepsilon_i$$

$$= \sum_{j=0}^{n} w_j x_{ij} + \varepsilon_i \quad \Big| \quad x_{i0} = 1$$

For hypothesis

$$h(x_i) = \hat{y}_i = w_0 + w_1 x_{i0} + w_2 x_{i2} + \cdots + w_n x_{in}$$

$$= \sum_{j=0}^{n} w_j x_{ij} \quad | \quad x_{i0} = 1$$

$$X_i = [x_{i0} \quad x_{i1} \quad x_{i2} \cdots \quad x_{in}]$$

Same thing in matrix form:

$$Y = XW + \mathcal{E}$$

m = training samples
n = dimension

dimension

$Y \rightarrow m \times 1$

$X \rightarrow m \times (n+1)$

$W \rightarrow (n+1) \times 1$

$\mathcal{E} \rightarrow m \times 1$

OLS estimation

$$W = (X^T X)^{-1} X^T Y$$

Observation (self-study)

- useful for small dataset
- require more computing for large dataset
- Problem when $(X^T X)^{-1}$ is non-invertible.

↓

① Perfect multicollinearity

② more predictor than observer.

Coefficient of determination

$$R^2 = 1 - \frac{SS_{Res}}{SS_{tot}}$$

SS_{Res} = Sum of squared residual = $\sum (y_i - \hat{y}_i)^2$

SS_{tot} = total sum of square = $\sum (y_i - \bar{y}_i)^2$

It is a statistical measure in linear regression that indicate how well the regression model explain the variability of dependent variable.

R^2 ranges 0 to 1

$R^2 = 1$ | The model is perfect fit ; explain all the variance.

$R^2 = 0$ | The model is poor fit.

Exercise

Estimate y using x by OLS considering the following dataset.

X	y
1	1·2
2	1·8
3	2·6
4	3·2
5	3·8

Estimate y for $x = 6$??

dismissed in class